AFRICAN STANDARD

ARS 1093

First Edition 2025

Butter — Specification

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This African Standard was prepared by ARSO Technical Committee on *Milk and Milk Products* (ARSO/TC 04).

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ARSO Central Secretariat
International House 3rd Floor
P. O. Box 57363 — 00200 City Square
NAIROBI, KENYA

Tel. +254-20-2224561, +254-20-3311641, +254-20-3311608

E-mail: arso@arso-oran.org
Web: www.arso-oran.org

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ARSO Central Secretariat
International House 3rd Floor
P.O. Box 57363 — 00200 City Square
NAIROBI, KENYA

Tel: +254-20-2224561, +254-20-3311641, +254-20-3311608

E-mail: arso@arso-oran.org
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Introduction

Butter is a fatty product derived exclusively from milk and/or products obtained from milk, principally in the form of an emulsion of the type water-in-oil.

This African Standard seeks to lay down requirements for the acceptable production of Butter in order to ensure quality, consumer safety and promote fair trade practices.

In elaborating this Standard, references were made to inputs from Manufacturers, National and International standards which are hereby duly acknowledged.

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Butter — Specification

1 Scope

This African Standard specifies requirements, sampling and test methods for butter, intended for direct consumption, or further processing.

2 Normative references

The following referenced documents are indispensable for the application of this Standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ARS 53**,** *General Principles of Food Hygiene – Code of practice*

ARS 56, *Pre-packaged Foods – Labelling*

ARS 1034*, Dairy industry — Glossary of terms*

ARS 1036*, Code of Hygienic Practice for Milk and Milk Products*

CXS 192, *General Standard for Food Additives*

CXS 193, *General Standard for Contaminants and Toxins in Food and Feed*

CXS 234, *Recommended methods of analysis and sampling*

CXS 346, General Standard for Labelling of non-retail containers of foods

ISO 707, *Milk and milk products — Guidance on sampling*

ISO 4833-1, *Microbiology of the food chain — Horizontal method for the enumeration of microorganisms Part 1: Colony count at 30 °C by the pour plate technique*

ISO 6579-1, *Microbiology of the food chain — Horizontal method for the detection, enumeration and serotyping of Salmonella — Part 1: Detection of Salmonella spp.*

ISO 6888-1, *Microbiology of the food chain — Horizontal method for the enumeration of coagulase-positive staphylococci (Staphylococcus aureus and other species) — Part 1: Method using Baird-Parker agar medium*

ISO 11290-1, *Microbiology of the food chain — Horizontal method for the detection and enumeration of Listeria monocytogenes and of Listeria spp. Part 1: Detection method*

ISO 14501, *Milk and milk powder — Determination of aflatoxin M1 content — Clean-up by immunoaffinity chromatography and determination by high-performance liquid chromatography*

ISO 16649-1, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of beta-glucuronidase-positive Escherichia coli — Part 1: Colony-count technique at 44 degrees C using membranes and 5-bromo-4-chloro-3-indolyl beta-D-glucuronide*

ISO 21528-2, *Microbiology of the food chain — Horizontal method for the detection and enumeration of Enterobacteriaceae Part 2: Colony-count technique*

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3 Terms and definitions

For the purposes of this document, the terms and definitions given in ARS 1034 and the following shall apply.

3.1
butter
fatty product derived exclusively from milk and/or products obtained from milk, principally in the form of an emulsion of the type water-in-oil

4 Composition and requirements

4.1 Essential raw ingredients

Milk and/or products obtained from milk

4.2 Permitted ingredients

a) Potable water

b) Sodium chloride and/or other types of food grade salt

c) Starter cultures of harmless lactic acid and/or flavour producing bacteria

4.3 Compositional requirements

Butter shall meet the compositional requirements given in Table 1 when tested in accordance with the test methods specified therein. Butter may be classified as identified in Table 1 and meet the compositional requirements, as shown.

Table 1 — Compositional requirements for butter

Characteristics	Permissible limit, % (m/m)	Test method
Milk fat content, min	80,0	Refer to CXS 234
Water content, max	16,0	
Milk solids-not-fat content, max	2,0	

5 Food additives

Food additives may be used and only within the limits specified in CXS 192 - Category 02.2.1 (Butter).

6 Contaminants

6.1 Heavy metals

The products covered by this African Standard shall comply with those maximum limits for metal contaminants specified in CXS 193.

6.2 Aflatoxins

When tested in accordance with ISO 14501, the level of aflatoxin M1 shall not exceed 0.5 µg/kg.

6.3 Pesticide residues

Pesticide residue limits shall be in accordance with limits set by the Codex Alimentarius Commission for the product covered by this Standard.

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6.4 Veterinary drug residues

Veterinary drug residue limits shall be in accordance with limits set by the Codex Alimentarius Commission for the product covered by this Standard.

7 Hygiene

The products covered by this African Standard shall be produced, prepared and handled in accordance with the provisions of the appropriate sections of ARS 53 and ARS 1036.

Butter shall be free from microorganisms and products originating from microorganisms in amounts which may represent a hazard to human health.

Butter shall comply with microbiological limits given in Table 3 when tested in accordance with the test methods specified therein.

Table 3 – Microbiological limits for butter

Micro-organisms	Sampling plan		Limits		Test method reference	The stage at which the parameter is applied
	n	c	m	M		
Salmonella	5	0	Absence in 25g		ISO 6579-1	Final product during shelf-life
Listeria monocytogenes	5	0	Absence in 25g		ISO 11290-1	Final product during shelf-life
Coagulase-positive Staphylococci	5	1	10 cfu/g	10^2 cfu/g	ISO 6888-1 ISO 6888-2	Final product during shelf-life
E. Coli	5	2	10 cfu/g	10^2 cfu/g	ISO 16649-1 or -2	Final product during shelf-life
Enterobacteriaceae	5	2	10 cfu/g	10^2 cfu/g	ISO 21528-2	At the end of process
Total Aerobic mesophilic bacterial count	5	1	10^2 cfu/g	10^3 cfu/g	ISO 4833-1	During the process

Note:
n = number of units comprising the sample;
c = number of sample units giving values between m and M;
m= The level of the microbiological criterion required in the product;
M= Value or level of microbial limit not to be reached or greater than in any unit of the sample.

8 Packaging

Butter shall be packaged in suitable food grade containers which will safeguard the hygienic, nutritional, technological, and organoleptic qualities of the product during dispatch, transport and use of the product until the end of its shelf life.

9 Labelling

In addition to the provisions of ARS 56, the following specific provisions shall apply:

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9.1 Name of the food

(i) The name of the food shall be “Butter”. The name “butter” with a suitable qualification shall be used for butter with more than 95% fat.

(ii) Butter may be labelled to indicate whether it is salted or unsalted according to national legislation.

9.2 Declaration of total fat content

If the consumer would be misled by the omission, the milkfat content shall be declared in a manner found acceptable in the country of sale to the final consumer, either

(i) as a percentage by mass, or

(ii) in grams per serving as quantified in the label provided that the number of servings is stated.

9.3 Labelling of non-retail containers

Information required in clause 8 of this Standard needed to be complied with the requirements mentioned in CXS 346.

10 Methods of sampling

For checking the compliance with this standard, the methods sampling contained in ISO 707, shall be used.

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Bibliography

CXS 279-1971 Standard for butter

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